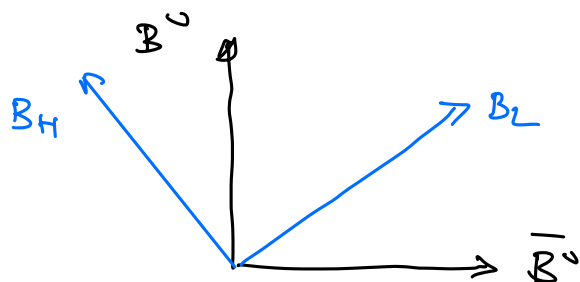


B^0 oscillations

$$B^0 \longrightarrow \bar{B}^0 \longrightarrow B^0$$

$B^0, \bar{B}^0$ flavor eigenstates $\neq$ Hamiltonian eigenstates



In absence of CP in mixing/oscillation

$$\begin{aligned} |B^0\rangle &= \frac{1}{\sqrt{2}} (|B_H\rangle + |B_L\rangle) \\ |\bar{B}^0\rangle &= \frac{1}{\sqrt{2}} (|B_H\rangle - |B_L\rangle) \end{aligned} \quad \Rightarrow \quad \begin{aligned} |B_H\rangle &= \frac{1}{\sqrt{2}} (|B^0\rangle + |\bar{B}^0\rangle) \\ |B_L\rangle &= \frac{1}{\sqrt{2}} (|B^0\rangle - |\bar{B}^0\rangle) \end{aligned}$$

$$|B_H, t\rangle = e^{-im_H t} e^{-\frac{\Gamma}{2} t} |B_H\rangle$$

$$|B_L, t\rangle = e^{-im_L t} e^{-\frac{\Gamma}{2} t} |B_L\rangle$$

$$\Gamma_H = \Gamma_L = \frac{1}{\tau}$$

$$|B^0, t\rangle = \frac{1}{\sqrt{2}} (|B_H, t\rangle + |B_L, t\rangle) =$$

$$= \frac{e^{-\frac{\Gamma}{2} t}}{\sqrt{2}} \left(e^{-im_H t} |B_H\rangle + e^{-im_L t} |B_L\rangle \right)$$

$$= \frac{e^{-\frac{\Gamma}{2} t}}{\sqrt{2} \sqrt{2}} \left(e^{-im_H t} (|B^0\rangle + |\bar{B}^0\rangle) + e^{-im_L t} (|B^0\rangle - |\bar{B}^0\rangle) \right)$$

$$= \frac{e^{-\frac{\Gamma}{2} t}}{2} \left((e^{-im_H t} + e^{-im_L t}) |B^0\rangle + (e^{-im_H t} - e^{-im_L t}) |\bar{B}^0\rangle \right)$$

$$\Delta m =: m_H - m_L > 0$$

$$m_H = m + \frac{\Delta m}{2}$$

$$m = \frac{m_H + m_L}{2}$$

$$m_L = m - \frac{\Delta m}{2}$$

$$= \frac{e^{-\frac{\Gamma}{2}t}}{2} \left[\left(e^{-imt} e^{-i\frac{\Delta m}{2}t} + e^{-imt} e^{+i\frac{\Delta m}{2}t} \right) |B^0\rangle \right. \\ \left. + \left(e^{-imt} e^{-i\frac{\Delta m}{2}t} - e^{-imt} e^{+i\frac{\Delta m}{2}t} \right) |\bar{B}^0\rangle \right]$$

$$= \frac{e^{-\frac{\Gamma}{2}t} e^{-imt}}{2} \left[2 \cos \frac{\Delta m}{2} t |B^0\rangle + \right. \\ \left. - 2i \sin \frac{\Delta m}{2} t |\bar{B}^0\rangle \right]$$

$$P(B^0 \rightarrow B^0) = |\langle B^0 | B^0(t) \rangle|^2$$

$$\cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}$$

$$= e^{-\Gamma t} \cos^2 \frac{\Delta m}{2} t$$

$$= e^{-\Gamma t} \left[\frac{1 + \cos \Delta m t}{2} \right]$$

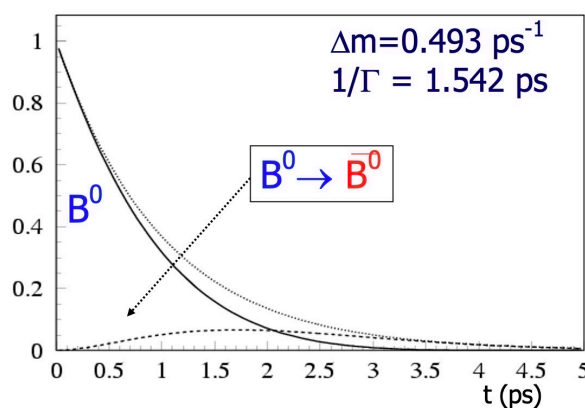
$$= \frac{e^{-\Gamma t}}{2} [1 + \cos \Delta m t]$$

Prob of
non oscillation
/ survival.

$$P(B^0 \rightarrow \bar{B}^0) = |\langle \bar{B}^0 | B^0(t) \rangle| \propto \sin^2 \frac{\Delta m}{2} t$$

$$= \frac{e^{-\Gamma t}}{2} [1 - \cos \Delta m t]$$

prob of
oscillation.



most generic case

$$|B^0\rangle = p|B_H\rangle + q|B_L\rangle$$

$$|p|^2 + |q|^2 = 1$$

$$|\bar{B}^0\rangle = p|B_H\rangle - q|B_L\rangle$$

$$P(B^0 \rightarrow B^0) = \frac{1}{2} e^{-\Gamma t} \left| \frac{q}{p} \right|^2 (1 + \cos \Delta m t)$$

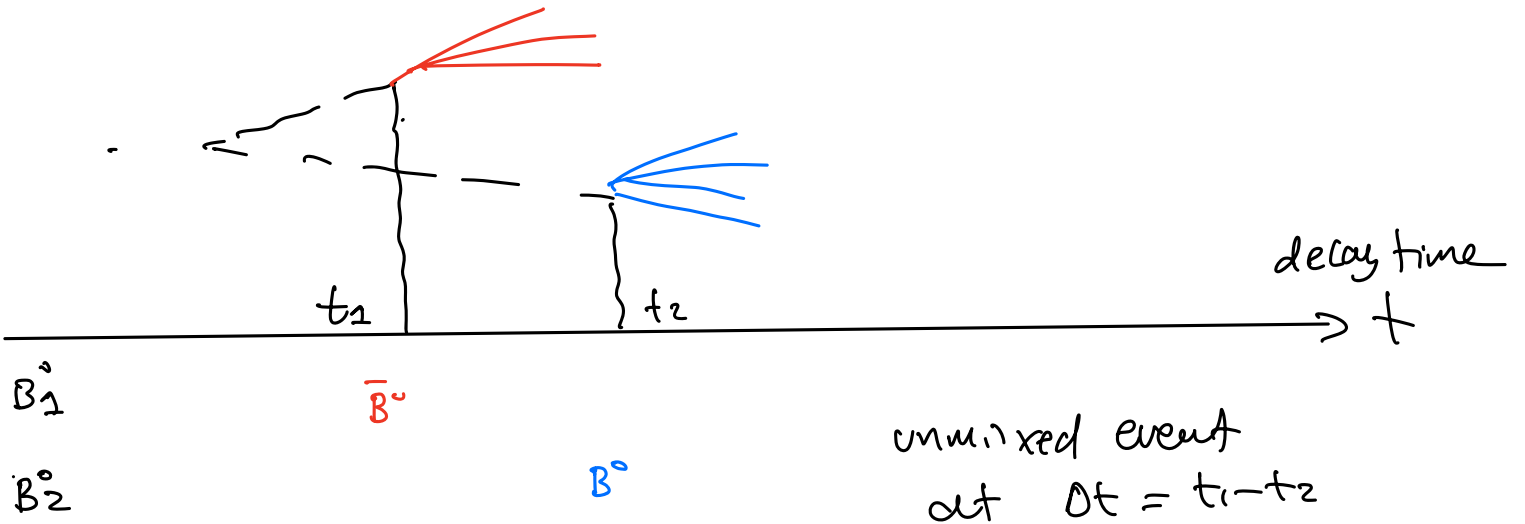
$$P(B^0 \rightarrow \bar{B}^0) = \frac{1}{2} e^{-\Gamma t} \left| \frac{q}{p} \right|^2 (1 - \cos \Delta m t)$$

① $t=0$ $B^0, \bar{B}^0$

② some time $t > 0$ $t_1, t_2 > 0$

mixed events $\# B^0 B^0 + \# \bar{B}^0 \bar{B}^0$ oscillation

unmixed events $\# B^0 \bar{B}^0 + \# \bar{B}^0 B^0$ no oscillation.



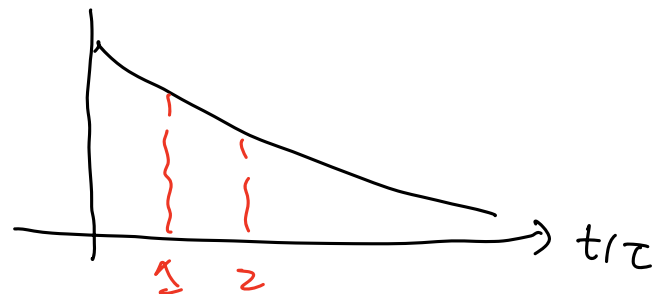
$\bar{B}^0 B^0$ observed

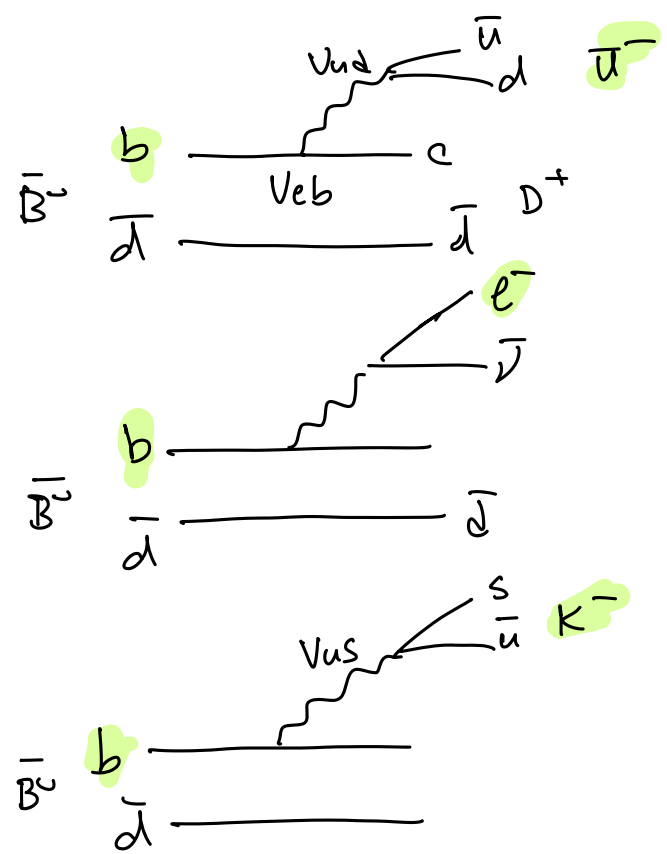
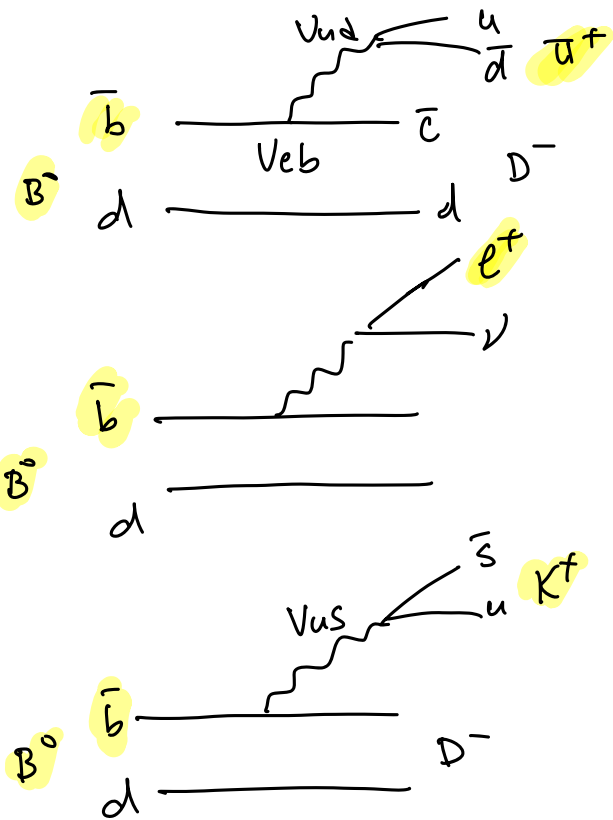
1) no mixing

2) $B^0 \rightarrow \bar{B}^0, \bar{B}^0 \rightarrow B^0$

3) $B^0 \rightarrow \bar{B}^0 \rightarrow B^0, \bar{B}^0$

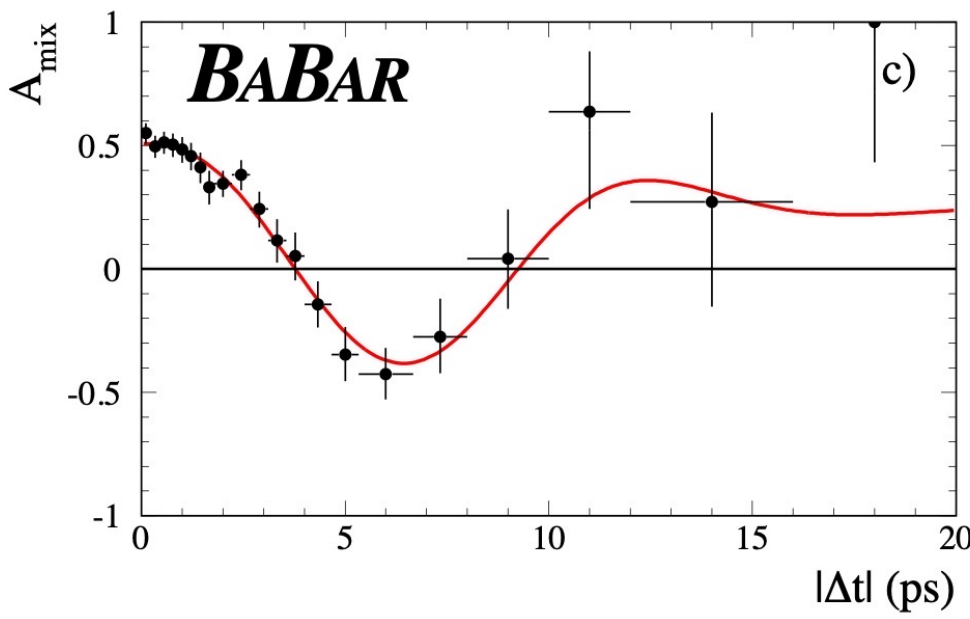
$$N(t) = N_0 e^{-\Gamma t} = N_0 e^{-t/\tau}$$





Flavor tagging: estimate flavor of original b quark from charge association.

$$A_{\text{mix}} = \frac{\#_{\text{mixed}} - \#_{\text{unmixed}}}{\#_{\text{mixed}} + \#_{\text{unmixed}}} \propto \cos \Delta m t$$



$$N(t=20 \text{ ps}) = N_0 e^{-\frac{20 \text{ ps}}{1.5 \text{ ps}}}$$

Entangled State

$$|B_1^0 B_2^0\rangle = \frac{1}{\sqrt{2}} (|B^0\rangle |\bar{B}^0\rangle - |\bar{B}^0\rangle |B^0\rangle)$$

if $t_B \rightarrow \infty$ always have $\frac{1}{2} B^0, \frac{1}{2} \bar{B}^0$

use hamiltonian eigenstates

$$|\psi(t=0)\rangle = \frac{1}{\sqrt{2}} (|B_H\rangle_1 |B_L\rangle_2 - |B_L\rangle_1 |B_H\rangle_2)$$

$$|\psi(t_1, t_2)\rangle = \frac{1}{\sqrt{2}} \left(e^{-i m_H t_1} e^{-i m_L t_2} |B_H B_L\rangle + e^{-i m_H t_2} e^{-i m_L t_1} |B_L B_H\rangle \right)$$

t_1 : time of observation of flavor of B_1

t_2 : time " " " " " " B_2

$t_1 = t_2 = t$ look at them at the same time.

$$= \frac{e^{-i(m_H + m_L)t}}{\sqrt{2}} (|B_H B_L\rangle - |B_L B_H\rangle)$$

$$B_H = \frac{1}{\sqrt{2}} (B^0 + \bar{B}^0) \quad B_L = \frac{1}{\sqrt{2}} (B^0 - \bar{B}^0)$$

terms: $|B^0\rangle |B^0\rangle, |\bar{B}^0\rangle |\bar{B}^0\rangle$

$|B^0\rangle |\bar{B}^0\rangle, |\bar{B}^0\rangle |B^0\rangle$

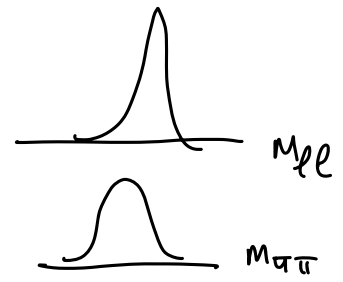
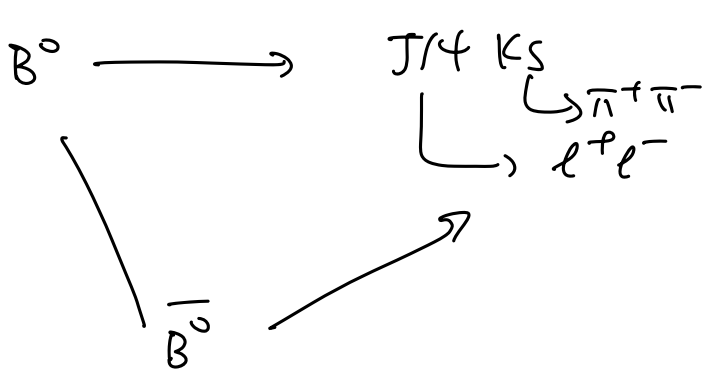
$$|\psi(t)\rangle = \frac{e^{-i(m_H + m_L)t}}{\sqrt{2}} (|B^0\rangle |\bar{B}^0\rangle - |\bar{B}^0\rangle |B^0\rangle)$$

pure state

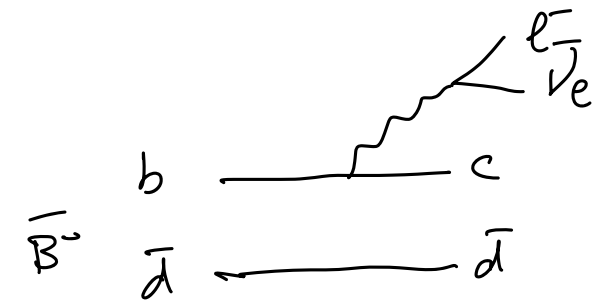
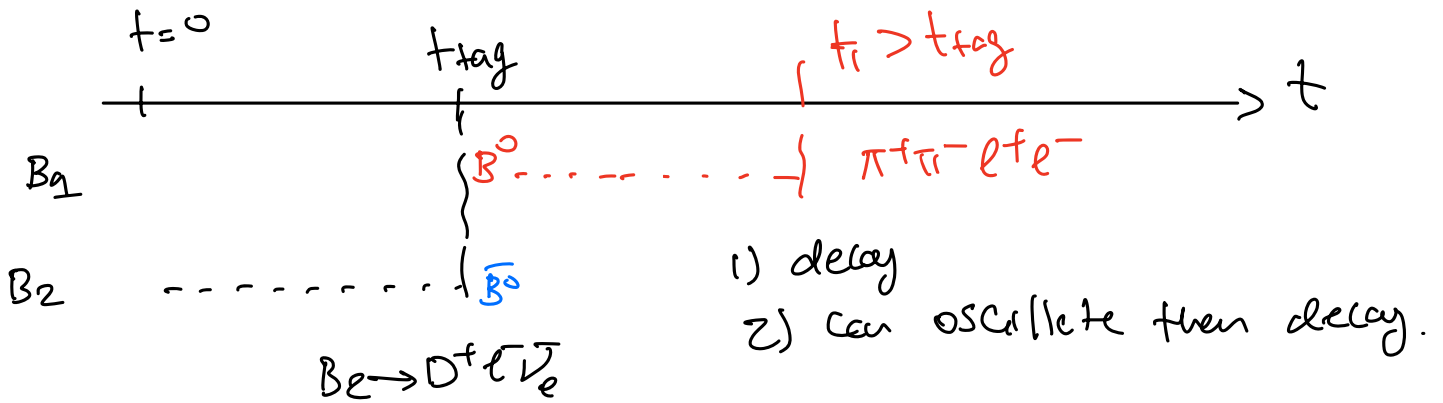
always $\frac{1}{2} B^0$

① all times $\frac{1}{2} B^0$

CP violation in interference between mixing and decay



B_2 delays at $t = t_{tag}$
 B_1 delays at $t = t_{cp}$
 $J/\psi K_S$ in the detector can be B^0 or $\bar{B}^0$



$$B_2 \equiv \bar{B}^0 \text{ @ } t = t_{tag}$$

$$B_1 \equiv B^0 \text{ @ } t = t_{tag} \text{ because of entanglement}$$

$$B_1 \equiv B^0 \text{ @ } t = t_{tag} \quad t_i > t_{tag} : \text{time of decay of } B_1$$

$$B_1 \rightarrow J/\psi K_S \rightarrow \pi^+\pi^- e^+e^- \text{ CP eigenstate.}$$

$$A(B^0_{tagged} \rightarrow J/\psi K_S, t_2 - t_{tag}) = A_1 + A_2$$

$$A_1: B^0 \rightarrow J/\psi K_S$$

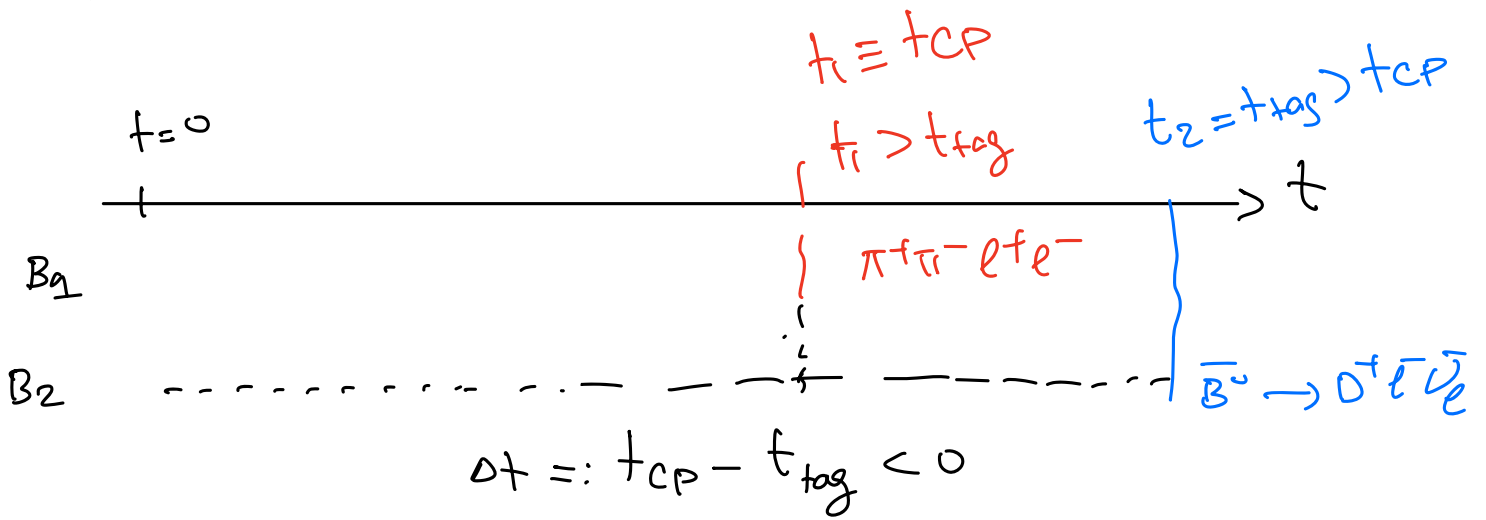
$$A_2: B^0 \rightarrow \bar{B}^0 \rightarrow J/\psi K_S$$

$$A = A_1 + A_2$$

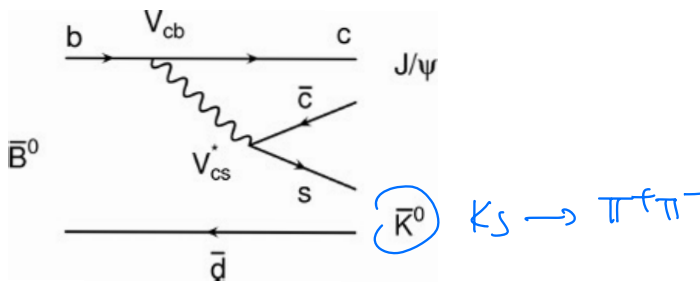
$$P(B_{tag}^0 \rightarrow J/\psi K_S, t) \propto |A_{tot}|^2 = |A_1|^2 + |A_2|^2 + A_1^\dagger A_2 + A_2^\dagger A_1$$

//

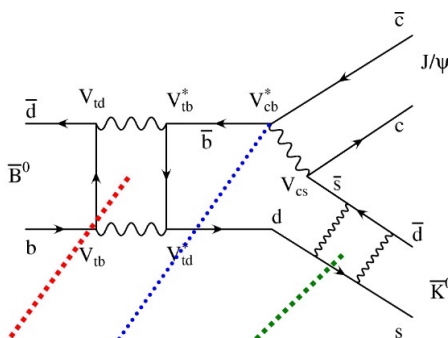
events



Flavor of B_1 and B_2 known at $t = t_{tag}$



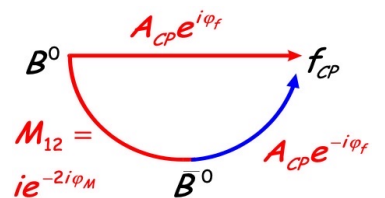
A_1 : delay.



A_2 : oscillate + delay

$K_S \rightarrow \pi^+ \pi^-$

$$\lambda_{\psi K_S} = \frac{q_B}{p_B} \frac{\bar{A}_{\psi K_S}}{A_{\psi K_S}} = - \frac{V_{tb}^* V_{td}}{V_{tb} V_{td}^*} \frac{V_{cb} V_{cs}^*}{V_{cb}^* V_{cs}} \frac{V_{cs} V_{cd}^*}{V_{cs}^* V_{cd}} = - \frac{V_{tb}^* V_{td}}{V_{tb} V_{td}^*} \frac{V_{cb} V_{cd}^*}{V_{cb}^* V_{cd}}$$



$$P(B_{CP} \text{ tagged as } \bar{B}^0, \Delta t) \propto \frac{e^{-\frac{\Delta t}{\tau}}}{4\tau} \left(1 + \sin(2\beta) \sin \Delta m \Delta t \right)$$

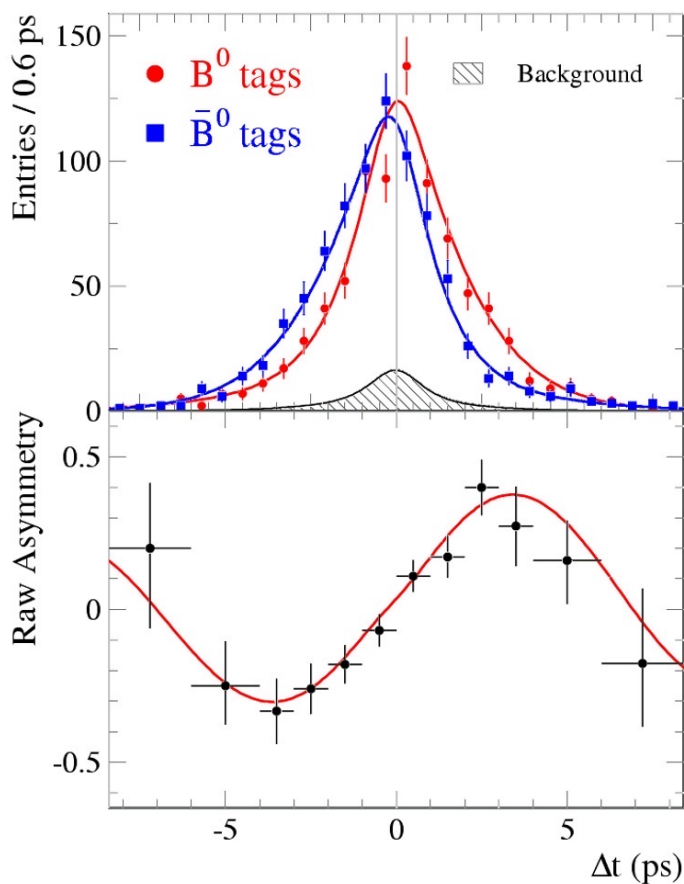
$$P(B_{CP} \text{ tagged as } B^0, \Delta t) \propto \frac{e^{-\frac{\Delta t}{\tau}}}{4\tau} \left(1 - \sin(2\beta) \sin \Delta m \Delta t \right)$$

count # J/ψ Ks events
as function of Δt

⇒ divide in 2 groups

- 1) tagged as B^0
- 2) tagged as $\bar{B}^0$

complex phase of CKM matrix



$$A_{CP} = \frac{\# B^0 \text{ tag} - \# \bar{B}^0 \text{ tag}}{\# B^0 \text{ tag} + \# \bar{B}^0 \text{ tag}}$$

$$\propto \sin(2\beta) \sin(\Delta m \Delta t)$$

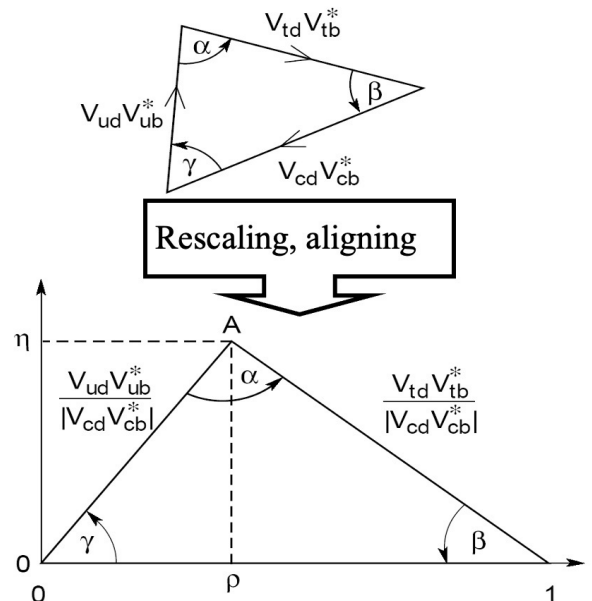
$$\propto \sin 2\beta$$

measured exp.
from mixing

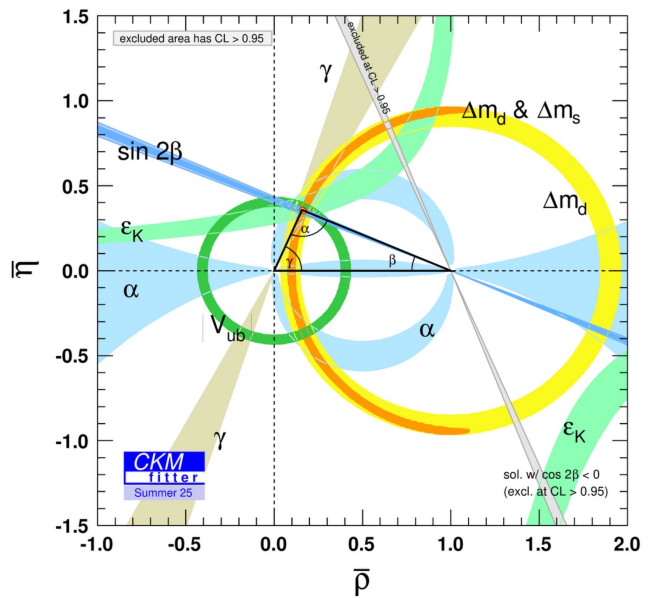
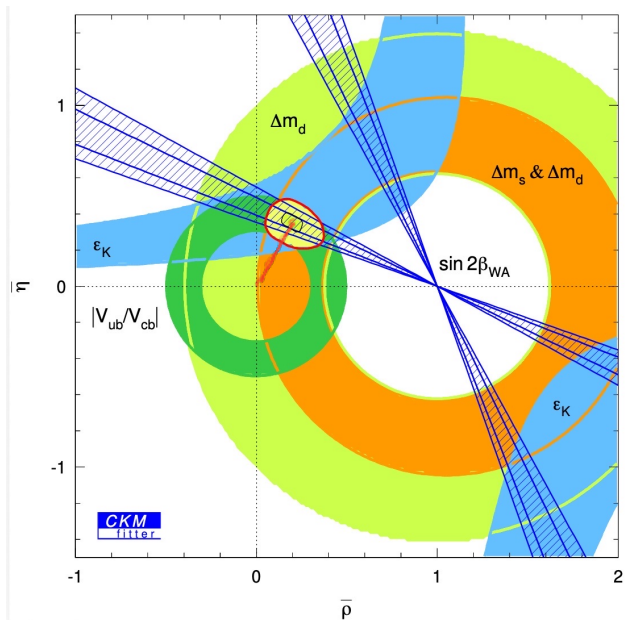
Summer 2002

$$\sin 2\beta = 0.741 \pm 0.067_{(stat)} \pm 0.033_{(syst)}$$

Evidence for CP
in mixing + decay.



Summer 2002



CP violation and CKM paradigm
proved experimentally

$\Rightarrow$ largely insufficient to explain
baryon asymmetry in the universe